

● LEXY

ENGINEERING DOCUMENTATION

DOCUMENT 09 / 09

Known Technical Debt & Backlog

AI Hiring Platform · Intelligence-First

Generated **June 14, 2026** · Confidential — internal use

Lexy — Known Technical Debt & Backlog

Engineering documentation set · Doc 9 of 9 What is intentionally deferred, and **why**. Debt is fine when it's a conscious choice with a recorded reason — this document is that record. Each item: **What** → **Why deferred** → **Impact** / **when to pay it down**.

■ A. Sourcing & enrichment

A1. `AbortSignal` is not propagated into provider adapters

- **What.** `runProvider` enforces a per-kind timeout (discovery 20s / enrichment 90s) so a hung provider can't block the sourcing fan-out — but on timeout it only *unblocks orchestration*; the in-flight network request keeps running until it finishes on its own.
- **Why deferred.** The timeout closed the user-visible bug (whole search hanging). Threading an `AbortSignal` through every adapter is a wider refactor with no additional *correctness* benefit today.
- **Impact / pay-down.** Wasted upstream work + a small window of orphaned sockets under heavy timeout rates. Pay down when adding more providers or when provider cost/latency metrics (Theme 2) show it matters.

A2. Legacy availability flags in `/sourcing/search`

- **What.** `pdlAvailable` / `serpAvailable` / `enrichLayerAvailable` flags predate the adapter registry and **don't reflect config-based disabling** (`SOURCING_DISABLED_PROVIDERS`).
- **Why deferred.** They're cosmetic/legacy; the registry's `skipped` reasons are the real source of truth now.
- **Impact / pay-down.** Mildly misleading status. Remove/redefine in terms of `isProviderEnabled()` next time the search response shape is touched.

A3. Enrichment seed URLs hardcoded to PDL + SERP

- **What.** The enrichment seeding step is wired to the pdl+serp pair rather than being generic over the provider registry.
- **Why deferred.** Those are the only enrichment providers today; generalizing before a second one exists is speculative.
- **Impact / pay-down.** Adding an enrichment provider requires touching seed logic, not just registering an adapter. Generalize when the 2nd enrichment provider lands (Roadmap Theme 5).

A4. `/sourcing/search` auth-hardening TODO

⚠ **Priority flag:** this sits low in the list but, because security ranks first (see Doc 4 Part D and Doc 6), it should be **scoped sooner than its placement suggests**. Do not read its position here as “low priority.”

- **What.** A noted auth-hardening follow-up on the `POST /sourcing/search` route. To be precise about scope: the route is **already authenticated** — it is *not* in the `withTenantContext` bypass list, so every call requires a valid HMAC-verified bearer token and runs under a resolved tenant context, and the body is Zod-validated. This is a **hardening gap, not an open/unauthenticated endpoint**.
- **What’s actually missing.** The follow-up is about **foreign-id authorization**: a caller supplies a `jobId` (and related ids), and those should be validated against `getAllowedTenantIds()` *before* the route links rows or fans the search out to external providers — the same standard applied elsewhere (see §C and the `tenant-scoped-foreign-ids` memory). Without it, the theoretical risk is a caller referencing an id outside their allowed tenant subtree on this specific route.
- **Impact if left.** Medium-bounded: exploitation requires an *already authenticated* user and is limited to cross-referencing ids within reach of the auth layer — not anonymous access — but it touches an external-spend, data- fan-out path, which is exactly where an authorization slip is most expensive.
- **Why deferred.** Tracked but not yet fully scoped; the surrounding tenant- gating standard (§C) already exists, so this is “apply the existing pattern to one more route,” not net-new design.
- **Pay-down.** Add an explicit `getAllowedTenantIds()` check on every caller-supplied id at the top of the handler, mirroring `/sourcing/nl-search` and the §C tenant-gating standard; add a test that a foreign `jobId` is rejected with `403`.

■ B. Observability

B1. AI/sourcing observability is logs-only

- **What.** Provider skip/timeout/error rates and scoring fallbacks are emitted as structured logs but **not** persisted to a metrics table or alerted on. (STT is the exception — it already persists `stt_transcribe_events` and alerts.)
- **Why deferred.** Logs were enough while autonomy was low and volume modest.
- **Impact / pay-down.** As autopilot/auto-scoring expand, “read the logs” stops scaling — degradations go unnoticed. This is **Roadmap Theme 2**; promote the STT health pattern into a general AI-ops surface.

B2. No health dashboard for the AI job queue / schedulers

- **What.** Queue depth/age/failure-rate and scheduler health aren’t surfaced.

- **Why deferred.** The queue (`SKIP LOCKED` , DB dedupe) is robust enough that it hasn't needed eyes yet.
- **Impact / pay-down.** A stuck queue or wedged scheduler is currently invisible until downstream symptoms appear. Pair with B1.

■ C. Tenant isolation hardening (ongoing discipline, not a one-time fix)

- **What.** RLS is a backstop; correct isolation still depends on every handler scoping with `getAllowedTenantIds` , gating foreign FKs, staff-allowlisting staff-only routes, and self-gating `WITH CHECK(true)` policies (`tps_insert` /push-to-client).
- **Why “deferred”.** This isn't a single ticket — it's a standing requirement that each new route must satisfy. The risk is *regression*, not an open hole.
- **Impact / pay-down.** Highest-severity class of bug if missed. Keep it in code review's #1 slot (Doc 6 §5); consider a lint/test that flags un-gated caller-supplied FKs. (*See Doc 4 Part B and the memory topic files.*)

■ D. Type system

D1. ~300 pre-existing `tsc` errors

- **What.** `tsc` reports ~300 errors, mostly Express request-param typing noise.
- **Why deferred (intentional).** The build gate is **esbuild**, not `tsc` . Driving `tsc` to zero is a large, low-value typing project that doesn't affect shipped behavior.
- **Impact / pay-down.** Editor noise + reduced type signal in affected handlers. Pay down opportunistically (e.g. typed Express request helpers) — not a blocking project. (*memory: express-params-tsc-noise*)

D2. `@workspace/db` must be rebuilt after schema changes

- **What.** After editing the Drizzle schema, `tsc` shows phantom “no exported member” until `@workspace/db` is rebuilt (`tsc -b`).
- **Why deferred.** Inherent to the project-references build setup.
- **Impact / pay-down.** A footgun for newcomers; documented in onboarding (Doc 3) and memory. Could be smoothed with a watch/prebuild step. (*memory: db-dist-typecheck*)

■ E. Interview pipeline residuals

- **What.** Real-time interview reliability is much improved (pooling, admission, watchdog, mobile STT, typing fallback) but remains the most environment-sensitive surface — most incidents were “fine in dev, broke under concurrency / on a specific device.”

- **Why deferred.** Each fix was reactive to an observed failure; full device/concurrency matrix coverage is open-ended.
- **Impact / pay-down.** Directly affects candidate conversion and brand. Tune pool sizing/admission from real metrics (depends on B1); broaden device coverage. (*Roadmap Theme 4.*)

■ F. Documentation & knowledge

- **What.** The deep “why” lives across `.agents/memory/` topic files; this engineering doc set is the first consolidated, downloadable digest.
- **Why deferred (until now).** Knowledge was captured incrementally in memory as it was learned.
- **Impact / pay-down.** Keep this set in sync — when a memory topic and these docs disagree, the memory files and code are newer; update the docs.

■ How this list is maintained

- A deliberate deferral gets an entry here **with its reason** — not a silent shortcut.
- When a code review surfaces a non-severe finding, it lands here rather than blocking the change (Doc 6 §5).
- Items graduate to the **Roadmap** (Doc 7) when they become active themes; the cross-references above show where each is already heading.

Note: two earlier follow-up tasks (referenced as #41/#42, around sourcing hardening) were **cancelled by the product owner** — their substance is captured in §A above so the reasoning isn’t lost even though the tickets are closed.